Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
C		2519836

Summary

Baron Coubertin once said ‘The important thing about the Olympics is not to win, but to participate; the essence of life is not to ask, but to strive.’ Our team developed a model based on the WMA-ARIMA algorithm that can reasonably predict the Olympic medal table:

For Task 1, after passing the significance test ($p<0.05$), we concluded that there is a linear relationship between the total number of medals (gold medals) of a country and the number of people sent by that country on each Sport. Based on the historical data, we used WMA and ARIMA models to predict the number of people and weights of each Sport that they might send in the 2028 Olympics, respectively, and then predicted the 2028 Olympic medal table through linear equations. Our model was 97.48% accurate in predicting the weights of the sports and a good fit was obtained in predicting the number of participants in 2024 (MSE=8.00). Based on the historical residual analysis, a reasonable prediction interval can be obtained by performing multiple Monte Carlo simulations. The ‘potential index’ of these countries was also calculated based on the similarity model with first-time winners, and the most appropriate threshold was fitted to 0.69, which predicted that four countries would win their first Olympic medals at the next Olympic Games, with an assessed odds ratio of 25%.

For task 2, we proved the existence of the ‘great coach’ effect by the Mann-Whitney U test ($p=0.0118<0.05$). By analysing the pearson coefficient (Pearson=-0.98) of the average weights before and after the great coach's coaching, we estimate that there is a fitable linear relationship between the effect of the great coach on Olympic performance. Based on the change of the weights of the events over the years, we choose Chinese athletics, Japanese women's volleyball team, and swimming of Cayman Islands as the events with potentials that are predicted to be increased by 22, 1, and 1 medals after the introduction of the great coach, respectively. 22, 1, 1 medals.

For Task 3, we compare the actual medals with the predicted medals after removing the ‘host’ effect, which proves the existence of the ‘host’ effect. Based on this, we believe that the host effect has an incremental impact on the number of U.S. medals.

We conduct a sensitivity analysis by randomly censoring the data, which proves that our model is very robust.

Finally, we find that the ratio of men to women participating in the Olympics is gradually approaching 1:1. for those countries that have never won a medal versus those that have won a medal for the first time in recent editions, they are geographically very close to each other, showing a concentrated distribution.

Overall, we have modelled the situation in relation to reality and completed a simple and practical, innovative and stable medal prediction model for the next Olympic Games.

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1 Introduction

1.1 Background

Since the opening of the first modern Olympic Games in Athens in 1896, more and more countries have participated in the Games, and the programs and connotations of the Olympic Games have been enriched with the advancement of time. With its long history, the Olympic Games have not only witnessed the limits of human physical movement, but also carried the vitality and vigor of the people of the global village across the centuries. The medal table of each Olympic Games attracts a lot of attention - to be able to have a place in the medal table represents the gradual progress of the country's sports level and sports ability, and to be able to be the host to host the Olympic Games is the supreme honor of a country. As a result, predicting the Olympic medal table has been a long-standing hot topic, and has become a trend in the reports of major data analytics media. However, the prediction of the number of medals that a country can win is usually not directly based on the change of the country's historical medal data, but rather on the strength of the athletes, etc., that the country will send to the upcoming games. In this thesis, we make predictions for the 2028 U.S. Olympic medal standings in Los Angeles based only on the athletes' historical program and medal data, as well as the historical medal standings and the historical host standings.

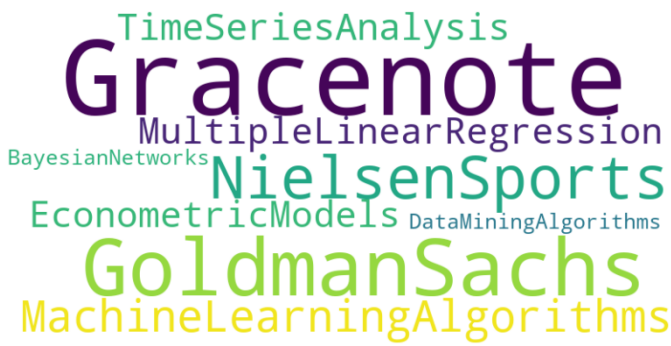


Figure 1: Internationally renowned sports forecasting agencies and their possible algorithms

1.2 Restatement of the Problem

- Build a model for **predicting medals**, using only the given data package,which includes:"summerOly athletes""summerOly hosts" "summerOly hosts""summerOly ho sts" the model should predict **the awards of each country in 2028**, including at least **the total number of medals** and **the number of gold medals** We will also give the changes in the number of medals for individual countries. number of medals and gold medals, and should also give the change in the number of medals for individual countries;.

- For countries that have never won a medal at the Summer Olympics, the model should

also make a reasonable prediction as to **whether or not** they will break through the history of zero medals to get their first Summer Olympics medal in 2028;

- Analyze the data to explore **the actual impact of the program** on the country's winning status, and find out the significance of the impact of a particular program on the winning status of a particular country by varying the model's parameters, etc;
- The "great coach" effect in Olympic history is very significant. We need to find out and verify the influence of "coaching effect" with the facts. Based on the above model, we can propose constructive training countermeasures for some countries, such as adding the "great coach" effect to a specific sport, which may help the country to break through the medal history in the Battle of Los Angeles.
- Propose other findings that can influence the number of medals in the modeling.

1.3 Our work

This is our work:

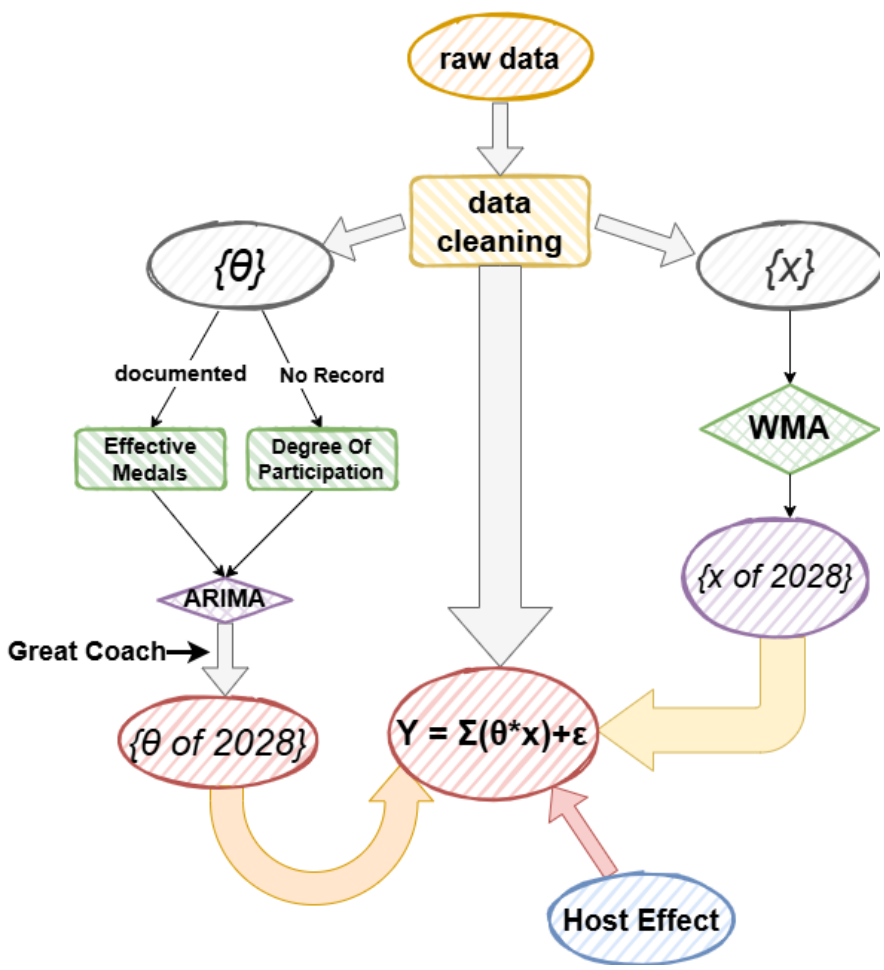


Figure 2: Our work frame

2 Assumptions and Assumption Testing

2.1 Assumptions

Assumption 1: Athletes (especially medalists) participating in the Summer Olympics in 2024 are near the normal distribution of individual performance trends[1], there is a probability that they will continue to win medals, or that other team members will "inherit" their medal-winning potential after their retirement. Overall, we assume that there will be no significant fluctuations in the number of participants in the 2028 Olympic Games.

Assumption 2: There will be no global emergencies in 2028 that would cause the Games to cease to be organized.

Assumption 3: All countries that participated in 2024 will continue to participate in 2028, and those that did not participate in 2024 will not continue to participate in 2028.

Assumption 4: No changes in 2028 except for additional projects.

2.2 Assumption Testing

Based on the processed data, we plot the change in the number of items set for the Olympics after 2000, as it shown in figure 2.

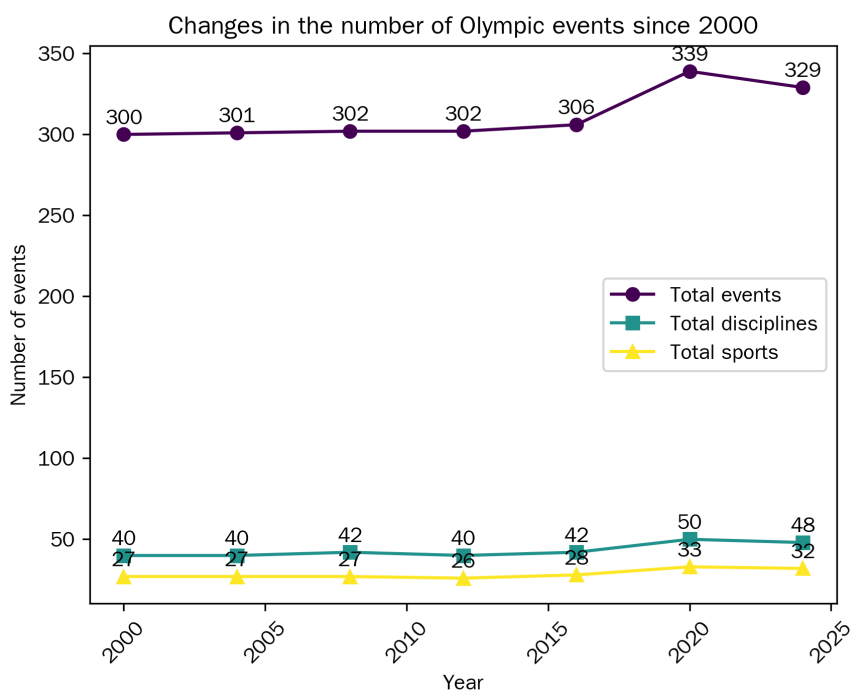


Figure 3: Changes in the number of Olympic events since 2000

We find that the number of participants in the Olympic Games after 2000 has gradually levelled off and that the types and number of sports set up by the Olympic Committee no longer change significantly. Therefore, we consider our fourth hypothesis to be reasonable.

3 Data Processing

3.1 Data Cleaning

For the given package, we first performed data cleaning. First we added missing values and filled in all items without medal data as "no medal". Next, we found some anomalies in the data:

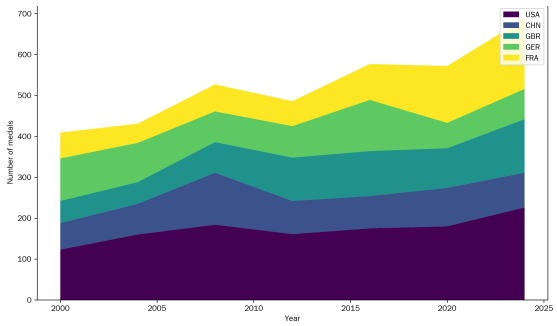
- In "sprots", "men" and "women" in "3*3baseketball" did not have any medals in some years. In the item "sprots", "men" and "women" in the item "3*3baseketball" did not have any medals in some years and were deleted.
- in the item "sprots", in the item "Athletics", the item "400 relay mixed" did not have any medals and was deleted.
- in the item "art competition", the item "unknown events" was not clear and was deleted.
- in the item "fencing", "brisket p 茅 e" was supposed to be a garbage code, but it was found to be the French word "épée", and it was corrected.
- in the item "NOC", "São Tomé and Príncipe" should be a garbled code, which was checked to be French "São Tomé and Príncipe". "is in Portuguese, and it has been modified.
- In the "athletes" file, we found that some athletes had two programs in "sports", such as "cycling road, triathlon". In the "athletes" file, we found that some athletes had two sports in "sports", such as "cycling road, triathlon", which we thought was due to the fact that they had entered two sports, and in order to avoid a mismatch between the "sports" and "programs" items, we split these items.
- In the "programs" file, we found that some of the items in "Discipline" had not appeared since 2000, so in order to improve the simplicity and accuracy of the model's data sources, we split the items that had not appeared before 2000 forward. In order to improve the simplicity and accuracy of the model's data source, we sorted forward and deleted any items that were not present before 2000.

3.2 Data Compression

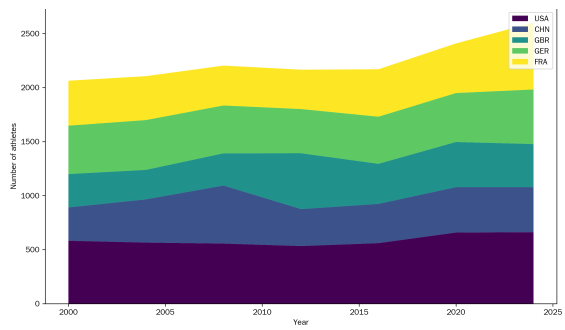
- We have merged the cleaned data 'Athletes' and 'Programs'.
- We categorised the data in Athletes by gender, and then compared and merged the data with the 'programme' in the same way as before.

3.3 Data Overview

We have collated the number of medals and the number of participants for each country over the years, as shown in the following two figures.



(a) Stacked Changes in the number of medals for selected countries since 2000



(b) Stacked Changes in the number of participating athletes for selected countries since 2000

Figure 4: The number of medals and competitors of each countries over years

4 Notations

Table 1: Notations used in our model

Symbol	Description
Total_Medal_i	Number of medals projected for country i
Gold_Medal_i	Number of gold medals projected for country i
$\text{Potential_Index}_n^i$	Potential capability that sport n will win awards for country i
Session_Similarity	The similarity between countries for sessions
Count_Similarity	The similarity between countries for number of people sent per session
θ_n^i	Weighting factor for country i on item n
θ_0^i	Weighting factors for country i prior to recruiting coaches
x_n^i	Number of participants sent by the country i on sport n
E_n	Number of events set up by sport n
ε	Impact of the remaining factors on the number of medals

5 Models

5.1 Solving of Problem 1

5.1.1 Model Construction

By studying and observing the data, we found that each item has a different impact on the award in each country, and in order to assess this impact, we introduced the weighting

factor $\theta \mathrm{x}_n$, which is calculated by the expressing formula:

$$\theta_n^i = \frac{\text{Number of medals won by country } i \text{ in sport } n}{\text{Total number of persons sent by country } i \text{ in sport } n} \tag{1}$$

In the above assumptions, we have found that the number of participants in the Olympic Games after 2000 has gradually levelled off, and the types and numbers of sports set by the Olympic Committee have not changed significantly. That is to say, it can be considered that the number of participants and awards of the Olympic Games are actually affected by the participation in recent years, so we use the Weighted Moving Average (WMA) algorithm combined with machine learning to predict x_n^i,as for θ_n^i,we use ARIMA to predict it, which can be assigned different weights according to the degree of importance of the data (usually near or far in time) by machine learning, and then calculate the weighted average. weights and then calculates the weighted average.

Now, we get a preliminary simple equation for predicting the number of medals can be obtained:

$$Total_Medal_i = \sum x_n^i \theta_n^i + \varepsilon \tag{2}$$

However, is it possible for a country that has never won a medal in its history to break through its zero medal history and get the country's first Olympic medal at the 2028 Summer Olympics? This would be a very important step forward for a country. We have also included this part of the calculation in our model.

We calculate the average number of sessions for first-time recipients: $First_Medal_Sessions_{ave}$ and average number of persons sent per session: $First_Medal_Person_{ave}$.

These two values serve as a reference for measuring the potential of countries that have never won a prize.

The countries that have never won a prize in 2024 are then compared to the first-time winners by calculating the similarity of the number of sessions and the number of people sent per session to the corresponding average values for the first-time winners.

$$\text{Session_Similarity} = 1 - \frac{|\text{Sessions} - \text{First_Medal_Sessions}_{\text{ave}}|}{\text{First_Medal_Sessions}_{\text{ave}}} \tag{3}$$

$$\text{Count_Similarity} = 1 - \frac{|\text{Per_Session_Count} - \text{First_Medal_Person}_{\text{ave}}|}{\text{First_Medal_Person}_{\text{ave}}} \tag{4}$$

We then compare the similarity of countries that have still never won a prize in 2024 with those that have won it for the first time:

The closer these two indicators are to 1, the closer the potential of these two countries to win the award is demonstrated. The similarity in the number of sessions sent and the similarity in the number of people sent per session were averaged to get the final 'potential index' :

$$\text{Potential_Index} = \frac{\text{Session_Similarity} + \text{Count_Similarity}}{2} \tag{5}$$

The average value is used because it combines the impact of the number of sessions and the number of persons sent per session on the potential to win the prize, which is considered to be equally important. The value of the potential index ranges from 0 to 1, with a value closer to 1 indicating that the country has a higher potential to win the prize in 2028. Finally, we have made a projection based on the number of first-time winners from 2000 to 2024, resulting in a list of first-time winners in 2028 and their winning programmes.

In summary, when predicting the number of medals for each country, we discuss the classification:

Country i's historical medal count is not 0: we use the WMA-Machine Learning algorithm to make predictions for θ and x respectively.

Country i's historical medal count is 0: we still use the same algorithm for x, while for θ we use the above comparison method to predict the potential index $Ptential_Index$, which is then assigned to θ.

For the interval of the predicted number of medals, we use the **Monte Carlo simulation algorithm**,That's because first, the residuals reflect the model error, and assuming that the future error pattern is consistent with the past, the historical residual distribution can be used to estimate the range of the future error; second, the use of empirical distributions avoids the assumption of a strong distribution and can reflect the nature of the error in a more realistic way; third, the Monte Carlo simulation samples from the empirical distribution of historical residuals and introduces uncertainty into the future predicted values; fourthly, the quantile is calculated by sorting the set generated by the simulation, and the confidence intervals of the predicted values can be estimated. This method has the obvious advantages of retaining data-driven flexibility, naturally quantifying uncertainty and being easily scalable to predict future values and uncertainty ranges.

The main idea of which is shown in the following:

Algorithm 1 Monte Carlo Simulation for Predicting Olympic Medal Count

Require: $T = \{2000, \ldots, 2024\}$, Y_{actual}, $Y_{predicted}$, Y_{2028}, $M = 10000$, $\alpha = 0.95$

$\mathcal{E} \leftarrow \emptyset$
for $t \in T$ **do**
  $\mathcal{E} \leftarrow \mathcal{E} \cup \{Y^t_{actual} - Y^t_{predicted}\}$
end for
$Y_{sim} \leftarrow \emptyset$
for $m = 1$ to M **do**
  $\epsilon \sim \mathcal{E}$; $Y_{sim} \leftarrow Y_{sim} \cup \{Y_{2028} + \epsilon\}$
end for
Sort Y_{sim}; $q_1 \leftarrow \frac{1-\alpha}{2}$; $q_2 \leftarrow \frac{1+\alpha}{2}$
$LowerBound \leftarrow \text{quantile}(Y_{sim}, q_1)$
$UpperBound \leftarrow \text{quantile}(Y_{sim}, q_2)$
return $[LowerBound, UpperBound]$

For the impact factor ε, it can be a combination of several factors:

- **'Host Effect':** As the 2028 Summer Olympics will be held in Los Angeles, USA, according to the document 'host' and the data of '2000-2024 National Games Participation and Medals', we filtered out the medal changes of the United States and the year it hosted the Olympic Games, using linear regression analysis to analyse and predict the 'host effect'. Based on the data of 'host' and [2000-2024 National Games Participation and Medal Count], we filtered out the medal changes of the United States and the year it hosted the Olympic Games, and used linear regression analysis to regress and predict the 'host effect'.
- **Policy and international situation:** For example, South Africa was denied participation in the Olympics after 1964 because of apartheid, and did not achieve results until 1992[2].
- **Physical condition of athletes:** The age at which athletes win awards may be consistent with "relative age effects"[3],which means the age distribution of national participating athletes may affect the awards.

  Since we can only use the given data in the predictive model, we **only use** the 'host effect' for the impact factor ε.

5.1.2 Model Rationality Analysis

As our proposed model points out that there is a linear relationship between the number of participants and the number of medals won, and it is a univariate linear relationship, we provide a plausibility analysis of the model. We attempt to construct a regression model to explore the relationships between relevant variables. The model is formulated as $Y = \sum_{i=1}^{n} \beta_i X_i + \epsilon$, where Y represents the total number of medals (obtained by summing up the medals of all major events), and X_i is the number of athletes in different sports (Sport)

for a particular country in a specific Olympic Games, serving as the independent variables in the regression analysis. The parameters β_i are the coefficients to be estimated and are constrained within the interval $[0, 1]$, and ϵ is the error term.

We make the following assumptions to ensure the rationality of the model: The error term ϵ has a mean of 0, i.e., $E(\epsilon) = 0$; its variance is a constant value σ^2, which means $Var(\epsilon) = \sigma^2$; and the error terms are independent of each other, expressed by the formula $Cov(\epsilon_i, \epsilon_j) = 0$ $(i \neq j)$.

Using data from multiple countries and multiple Olympic Games as multiple observation samples, we employ the constrained least - squares method to estimate the parameters β_i. Although the specific solution process is not presented here, a set of parameter solutions can be obtained, indicating that appropriate parameters can be found to fit the model to Y and $\{X_1, \cdots, X_n\}$.

Next, we conduct a regression significance test. The null hypothesis H_0 is: $\beta_1 = \beta_2 = \cdots = \beta_n = 0$, which implies that all independent variables have no significant linear effect on the dependent variable Y; the alternative hypothesis H_1 is: there exists at least one i such that $\beta_i \neq 0$, meaning that at least one independent variable has a significant linear relationship with the dependent variable Y.

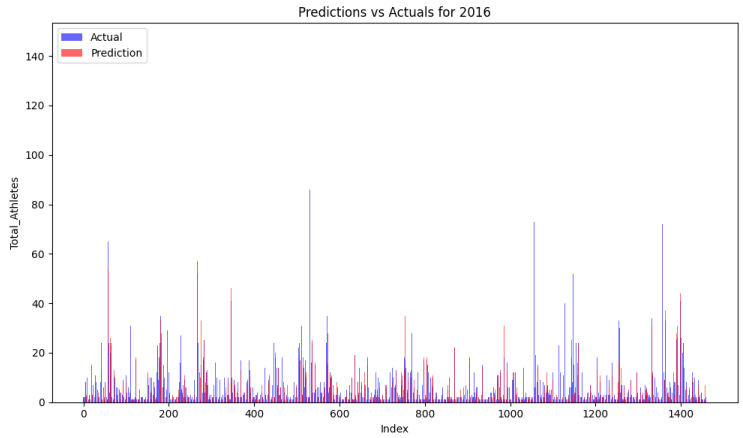
We perform the test by calculating the F - statistic. The formula for the F - statistic is $F = \frac{MSR}{MSE}$, where $MSR = \frac{SSR}{k}$ and $MSE = \frac{SSE}{n-k-1}$. Here, $SSR = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2$ is the regression sum of squares, which reflects the degree to which the independent variables explain the variation of the dependent variable; $SSE = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$ is the residual sum of squares, representing the part that the model cannot explain; k is the number of independent variables, and n is the sample size.

After calculation, we obtain F = **997.5** and the corresponding p - value is 8.72×10^{-3}. Since $p = 8.72 \times 10^{-3} < 0.05$, according to the significance test rules, we reject the null hypothesis $\mathbf{H}_0$. This indicates that the regression as a whole is significant, i.e., there is a significant linear relationship between the dependent variable Y and the independent variables X. Meanwhile, the coefficient of determination $R^2 =$ **0.747** of the model shows that **the independent variables can explain approximately** 74.7% **of the variation in the dependent variable, further validating the effectiveness of the regression model**.

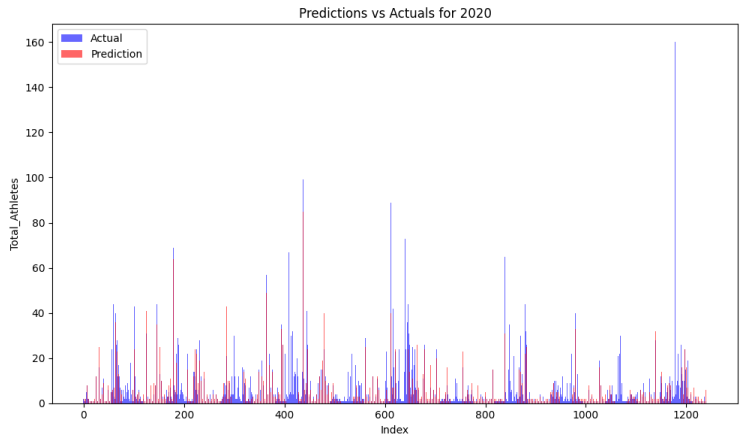
In conclusion, although an unconstrained ordinary least - squares method (OLS) was first used for an approximate test during the significance test, given that the OLS results show a significant relationship and most of the parameter estimates fall within the interval $[0, 1]$, it implies that the solution under the constrained conditions is also likely to be significant. Therefore, the regression model we constructed is reasonable and can well describe the linear relationship between the dependent variable Y and the independent variables X.

5.1.3 Model Solution

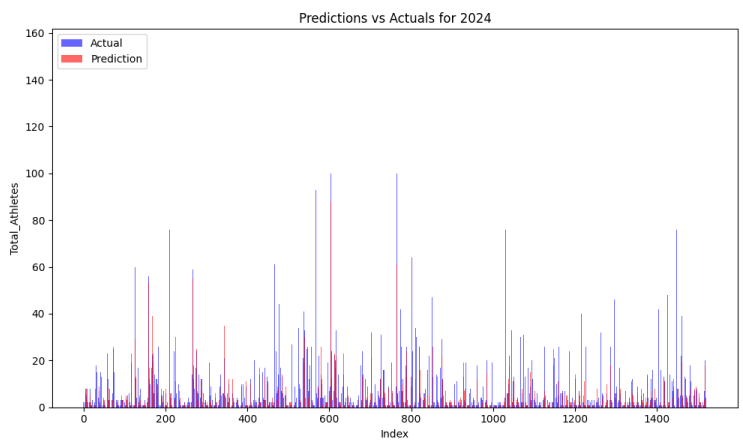
We compare our predicted data for x and θ with the actual data, and the figure below shows the comparison of the fitted data for each year:



(a) 2016



(b) 2020

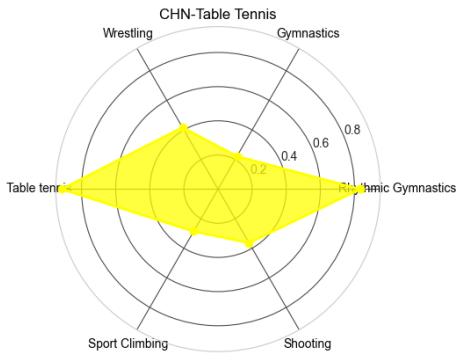


(c) 2024

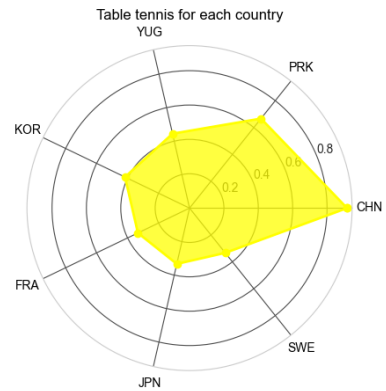
Figure 5: Comparison of predict and actual numbers of competitors, 2016-2024

For the calculated weighting factors θ, we compare them with the data of 2024, and the accuracy is 97.4%, and we take 2024 Chinese table tennis as an example to illustrate the

accuracy of its prediction:



(a) CHN-Tabletennis θ



(b) Table tennis θ for each country

Figure 6: two-dimensional comparison on θ

The radar chart in Figure (a) shows the weighting factor θ of the six events for China, which indicates that 'table tennis' contributes the most to China's medal count, with 'Rhythmic Gymnastics' coming in a close second; Figure (b) shows the extent to which 'table tennis' contributes differently for each country, and that only these countries have a projected θ for the event. This indicates that 'table tennis' contributes the most to China's medal count, with 'Rhythmic Gymnastics' following closely behind; Figure (b) shows the extent to which 'table tennis' contributes differently to each country, and that only these countries have a predicted θ value for the programme, the number of participants from these countries x is multiplied by the corresponding θ, and the medals are summed up. The number of medals for these countries x is multiplied by the corresponding θ, and the sum of these medals equals the total number of medals set for the event, which shows that our weighting factors not only take into account the local optimality, but also the overall reasonableness.

For the calculation of the 'host effect' ,we excluded the 'outliers' in the number of medals won by the United States in the past years (which may be caused by the 'host effect'), and then fitted a linear curve to the rest of the data, which resulted in a smooth curve.Next, we found out the data of the year when the U.S. hosted the Olympic Games, and made a difference with the linearly fitted data, and "Difference" is the difference value.

We input these differences into the above prediction model, make it relearn and adjust the weights, and finally obtain the impact factor corresponding to the 'Host Country Effect Index' predicted for 2028.

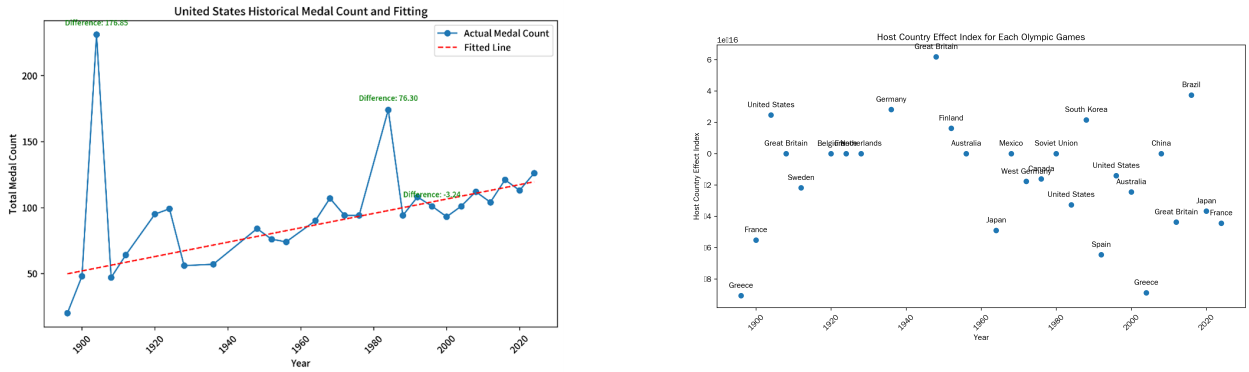


Figure 7: Fitted curve with ε scatter plot

Use the equation(2):

$$Total_Medal_i = \sum x_n^i \theta_n^i + \varepsilon$$

We have come up with a 2028 predicted medal counts, including the total medal counts and the gold medal counts, taking the **top 20** to be analysed and displayed:

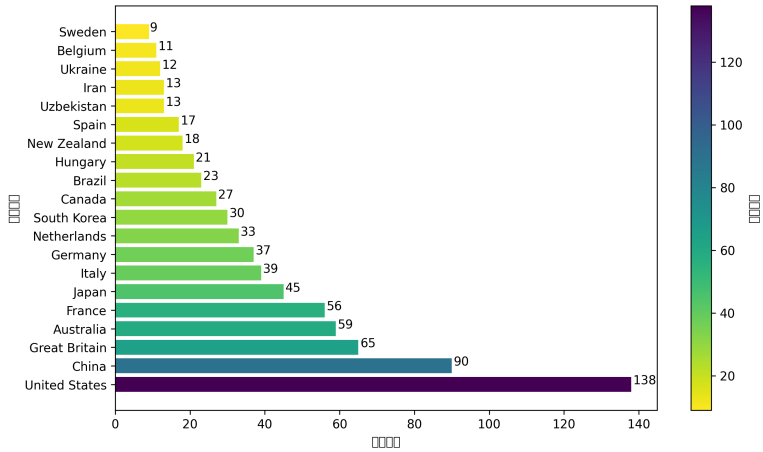


Figure 8: Total Medal Counts for 2028 TOP20 (predicted)

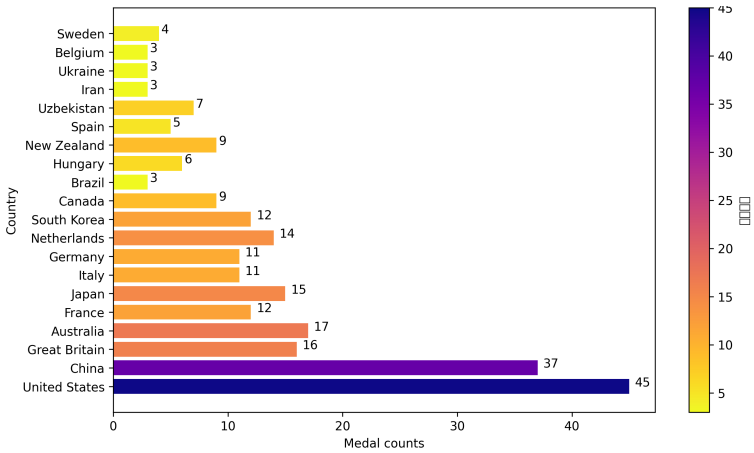


Figure 9: Gold Medal Counts for 2028 TOP20 (predicted)

Then we give a prediction range for the number of medals in the following table:

Table 2: Medal prediction ranges for the 2028 TOP20 countries(predicted)

NOC	Prediction Range of Total	Prediction Range of Gold
USA	133,146	42,49
CHN	84,94	36,39
GBR	61,69	16,18
AUS	55,62	15,19
FRA	51,57	11,15
JPN	42,50	14,18
ITA	37,41	7,15
GER	33,43	9,15
NED	31,35	13,16
KOR	28,36	11,15
CAN	22,32	8,9
BRA	18,29	1,5
HUN	16,23	6,9
NZL	14,23	8,11
ESP	14,21	3,6
UZB	12,18	7,10
IRI	12,18	2,5
UKR	8,15	1,5
BEL	10,14	3,4
SWE	8,10	4,5

As we can see, the difference between the medal table and last year's is mainly due to the fact that the United States has increased its number of medals thanks to the host effect, while France has lost the host effect with a corresponding decrease in the number of medals.

We vary the item weighting factor θ to explore the importance of the item for certain countries:

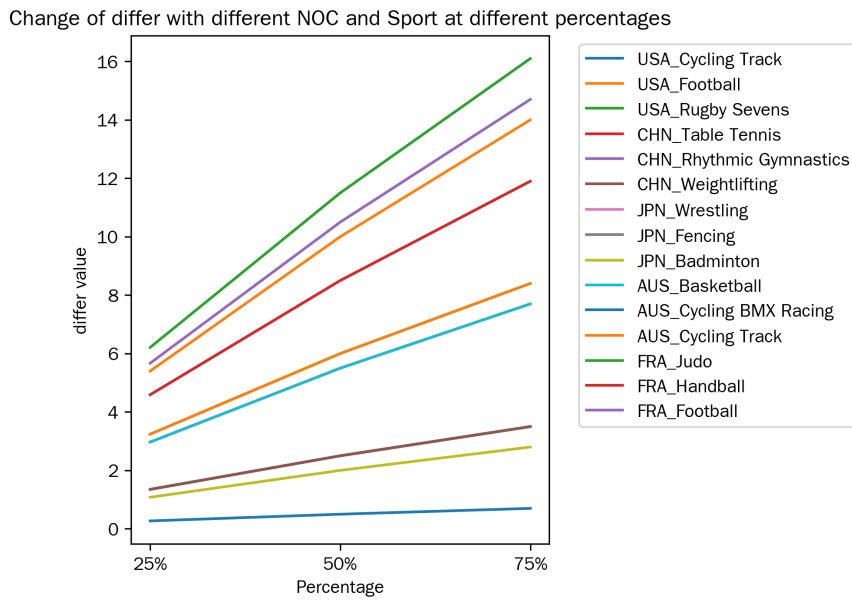


Figure 10: Change of differ with θ at 25% ,75% and 25%

In our model, the weighting factor θ of a project represents the importance of the project to that country.

We selected the weighting factor θ of a particular country for a particular item, subtracted its value by 50%, 25% and observed the change in the predicted number of medals, and found that the weighting factor and the number of medals of the country varied almost linearly, and the larger the weighting factor θ, the greater the impact of the item on the number of medals of a country.

Finally, we also calculated the potential index $Pontential_Index$ for countries that have never won a medal, and we believe that any country with a $Pontential_Index$ greater than **0.69** has a high probability of winning its first medal in 2028, so we predict that a total of four countries will break the zero-medal history in 2028:

We used $Pontential_Index$ against countries winning medals for the first time in 2016, 2020 and 2024 and calculated an accuracy rate of about **75%**, i.e. an error rate of about **25%**.

5.1.4 Analysis of Model Accuracy

For the error analysis of the model, we use **Mean Relative Error(MRE) the Mean Square Error (MSE)** and **the Mean Absolute Percentage Error (MAPE)** as a

Table 3: First-time-medal countries in 2028(predicted)

NOC	Sports	Pontential_Index
CAY	Swimming	0.8684
SKN	Athletics	0.7270
COD	Boxing	0.7242
BEN	Swimming	0.6913

measure, and since in our algorithm the weights are based on the data from the first three terms of the predicted year, we evaluated the prediction accuracy of the first three terms of the year 2028: 2016, 2020, and 2024.

Below is a 3D bar chart showing the change in error over years:

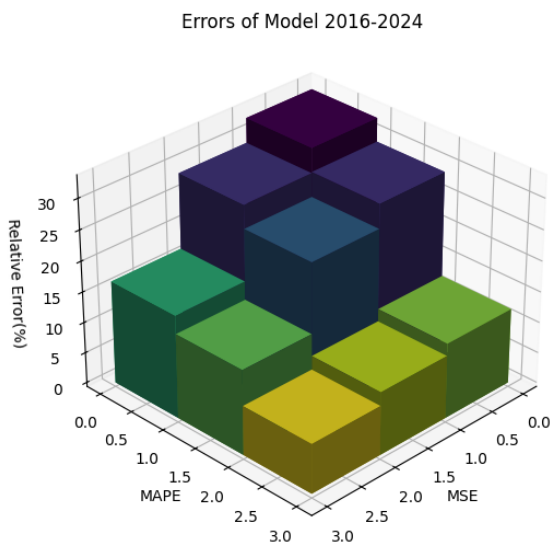
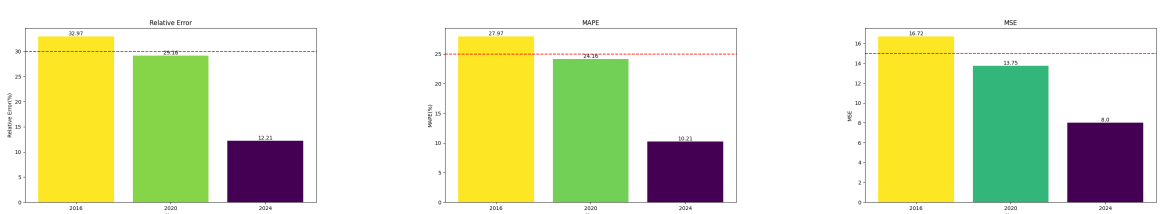


Figure 11: Errors of our model: MRE, MSE, MAPE



(a) Change in MRE (b) Change in MAPE (c) Change in MSE

Figure 12: Two-dimensional plot of changes in the three values over years

We categorised this Olympic medal forecast as a social science forecast and considered that data with MRE below **30%**, MAPE below **25%** and MSE below **15** fitted better, and based on the graph below, we found that our model evaluated the data to fit better, and the closer we got to the target year (2028), the better the prediction was.

5.2 Solving of Problem 2

5.2.1 Model Validation

The 'great coach' effect affects the weighting factor θ_n^i of the item to the country, and we verify the existence of the 'great coach' effect along the following statements:

Taking volleyball coach Lang Ping[4] as an example, Lang Ping started coaching the U.S. women's national volleyball team in 2005. In order to make the gold, silver and bronze medals equal, we make their share 3, 2 and 1 respectively the performance data of the US women's volleyball team before and after Lang Ping's coaching is divided into 10 sessions of the Olympic Games, and it is found that the level of the women's volleyball team before and after Lang Ping's coaching has changed as shown in the **table 2**.

Table 4: U.S. volleyball team Medal Counts and processed Numeric Value, 1988-2024

Year	**Medal_Counts**	**Numeric_Value**
1988	NO MEDAL	0
1992	BRONZE	1
1996	NO MEDAL	0
2000	NO MEDAL	0
2004	NO MEDAL	0
2008	SILVER	2
2012	SILVER	2
2016	BRONZE	1
2020	GOLD	3
2024	SILVER	2

Considering that the two sets of data before and after are not repeated measurements of the same team (or the same game) in two different states, but rather two sets of observations from two different time periods, the paired test is not appropriate; the t-test is also inappropriate due to the difficulty of satisfying the Central Limit Theorem with a sample size too small and the fact that the data are essentially discrete/ordered variables, where the assumption of normality is more difficult to be established; chi-square tests are usually used with frequency data or series tables, whereas here it is more straightforward to compare two sets of continuous (discrete scoring but can be treated as continuous) data, so the chi-square

test is also inappropriate. Due to the small sample size (5 observations each) and the fact that the data itself is non-normal, discrete/ordered, it would be more appropriate to use a non-parametric test to compare the two sets of data to see if there is a significant difference.

Mann-Whitney U test, also known as **the Wilcoxon rank sum test** belongs to the non-parametric test methods, used to compare the distributional differences between two groups of independent samples (usually can be regarded as the median or distributional position differences), does not require the sample data to meet the normal distribution. The basic idea is to mix the two sets of data after ranking, according to the rank assigned to the sample rank (rank), and then count the difference between the rank sum of the two sets of data, and calculate the corresponding test statistic U and its p-value.

We used the Mann-Whitney U test to verify that there was a significant difference between the before and after data: ordering a significance level of $\alpha = 0.05$:

Original Hypothesis (H0): there is no significant difference in position between the two data sets.

Alternative hypothesis (H1): there is a significant difference.

Rejection domain:{p|p<0.05}

p=0.0118<0.05 falls into the H0 rejection domain

Therefore, we reject the original hypothesis at the **5%** level of significance and conclude that there is a statistically significant difference between the two data sets of ‘before and after coaching results’. This proves that there is a coaching effect.

5.2.2 Model Construction

After conducting in - depth verification and analysis of numerous datasets on the medal - winning situations of various countries, we focused on the impact of great coaches on athletes' performan[5]. By accurately calculating the project weights and plotting the trend graph of these weights over the years, we identified two other highly representative great coaches.

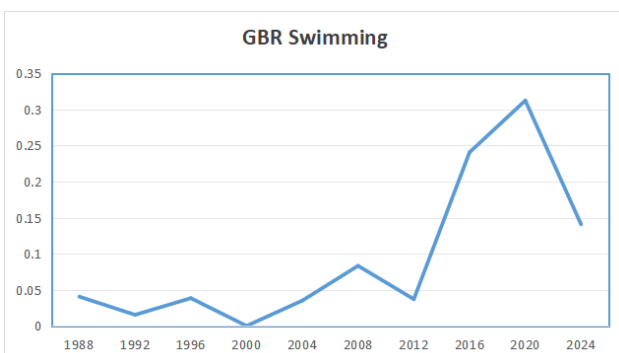


Figure 13: θ of swimming changes in GBR

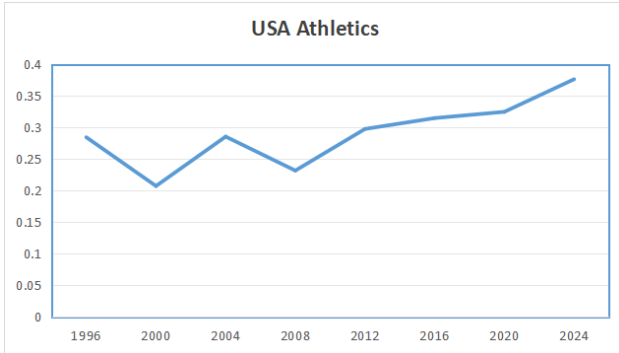


Figure 14: θ of athletics changes in USA

The first is Alberto Salazar[6] from the United States. Since 2012, he has been coaching the long - distance running events in the United States. It can be clearly observed from the line graph of the effective ratio that before and after his coaching, the weight of the long - distance running events in the United States showed a relatively stable upward trend. BillSweetenha[7], the head coach of the British swimming team. He has completely revamped the training system of British swimming. After 2012, the medal - winning ratio of the British swimming team increased rapidly, and the change in performance was quite remarkable.

To quantify the impact of coaches on the project weights, we conducted a comparative analysis of the effective ratios of the corresponding projects in relevant countries before and after the coaches took charge. The research findings indicate that the change in the effective ratio follows the formula: $\Delta\theta = a \cdot \theta_0 + b$.

Based on this, we calculated the relevant parameters $\langle a, b \rangle$ for the corresponding sports under the coaching of these three coaches, respectively. These parameters will serve as the basis for calculating the expected medal - winning results when providing recommendations to relevant countries in the future.

Considering the outstanding achievements of Lang Ping, the coach of the Chinese women's volleyball team, as mentioned above, we comprehensively took into account various factors such as the characteristics of sports, the geographical locations of countries, sports policies, and the physical qualities of athletes. We screened out three countries and believe that these countries can enhance the medal - winning ratios of relevant sports by hiring excellent coaches, thus having the potential to win medals in events like the Olympic Games. The following is a detailed discussion one by one.

5.2.3 Model Solution

We have selected three representative countries to give our analyses and recommendations:

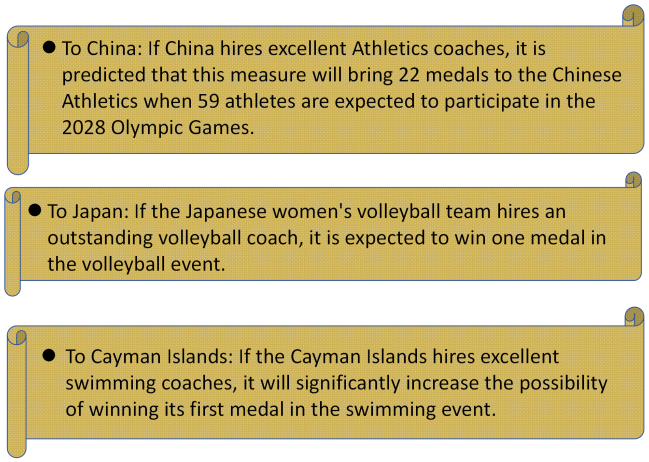


Figure 15: Our suggestions

5.3 Solving of Problem 3

Throughout the history of the Olympic Games, men have always dominated, as shown in the posters for the 1920, 1924 and 1988 Olympic Games, which clearly have a male element to them.

We have analysed the changes in the gender ratio of the Summer Olympics from 1896 to 2024:

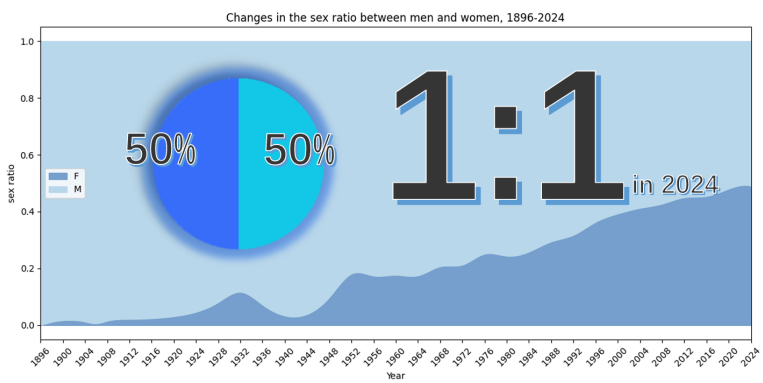


Figure 16: The sex ratio changes 1896 to 2024

And we found that the proportion of female participation is basically in an upward state from about 1940 onwards, which indicates that women's participation in the Olympic Games has been increasing from year to year, which is not only related to the progress of the society, but also the awakening of women's power. According to the prediction of our model, if the proportion of female athletes becomes larger, the number of female athletes participating in the Olympic Games will also increase, given the same level of increase in the number of participants.

When we make projections for non-winning countries, we find similarities in the geographic distribution of non-winning countries and first-time medallists (blue dots indicate first-time winners up to 2024, and red dots indicate countries that have never won a medal up to 2024):

Figure 17: Geographical distribution of sports-poor countries

They are concentrated in Africa, South America and Oceania, which suggests that countries in these regions may not have had the opportunity to participate in the Olympics, or to be better trained. So we think the **"great coach"effect could make a big difference when it comes to these countries**.

6 Sensitivity Analysis

Prediction for x: The extent to which the accuracy of the model's predictions for the year 2024 decreases by removing certain data:

Analytical Solution: We deleted 10%, 20%, and 30% of the data in the dataset, and then used the weighted moving average algorithm to predict the number of participants in each project in 2024, and compared it with the actual value, and used MSE to indicate the degree of error. It can be seen that when 30% of the data is deleted, the value of MSE is still around 15, indicating that the model is relatively robust.

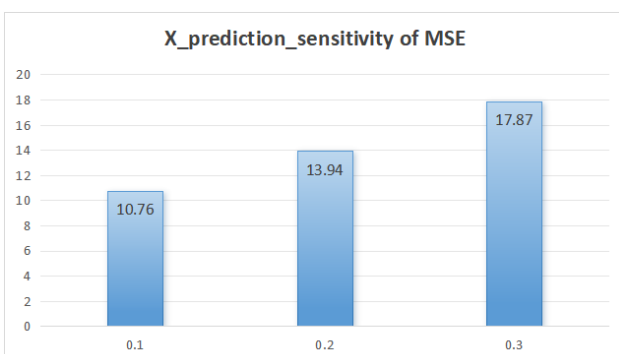


Figure 18: sensitivity of x

For the prediction of θ: The judgement method is the same as above.

Analysis scheme: 10%, 20%, and 30% of the data in the data set were deleted, and then the algorithm combining ARIMA and time series was used to predict the project weights of each project in 2024, 2020, 2016, and 2012, and compared with the actual values, the average MSE value under the same deletion ratio was calculated, and the MSE was used to indicate the degree of error. It can be seen that under the premise of missing data, the value of MSE is stable at about 3.65, indicating that the model prediction is accurate and relatively robust.

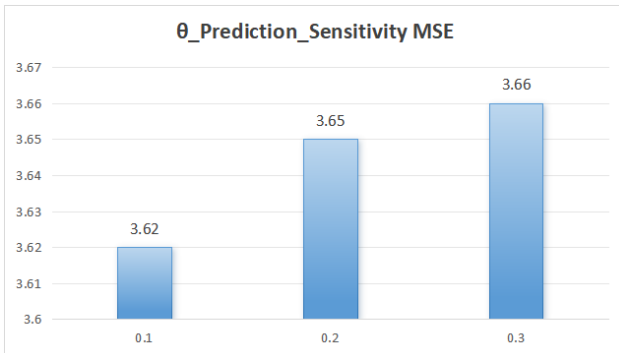


Figure 19: sensitivity of θ

7 Model Evaluation

7.1 Model Evaluation

7.1.1 Advantages

In the process of modeling, we took into account textbfthe logistic relationship between the independent and dependent variables and passed the significance test, making the model sufficiently reasonable.

And the model is not superficially time-series forecasting by the number of medals of a country in the past, but takes into account the fact that the number of medals of a country is accumulated by the number of medals of all the Sports. At the same time, the country's medal count in a Sport is reasonably subdivided into the product of the weight of the event and the number of athletes sent in that Sport.

A lot of thought has gone into the definition of θ, so that theta is **perfectly adapted** not only to countries that have a record of winning medals, but also to countries that have never won a medal. This allows us to predict medals for all countries in the world with a record of participation.

7.1.2 Limitations

Due to the limitations of the available data, the model fails to include all potential influences on a country's medal count, such as GDP, national sports culture, etc.

8 Conclusions

This research centers on predicting the 2028 Olympic medal standings, developing a comprehensive and effective model. By analyzing historical data and using methods like the Weighted Moving Average and ARIMA models, it accounts for key factors such as the number of participants, project weights, the ”great coach” effect, and the ”host” effect. The model predicts countries’ total medals, gold medals, and their trends, and analyzes medal - winning chances for countries with no prior medals. It shows excellent performance in predicting 2024 data, with a 97.4Moreover, the research reveals that the male - female participation ratio in the Olympics is becoming more balanced, and non - medal - winning and first - time medal - winning countries are geographically concentrated. Despite limitations from existing data, lacking factors like GDP and sports culture, the model offers new perspectives and scientific methods for medal standings prediction, which is valuable for countries’ sports planning.

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